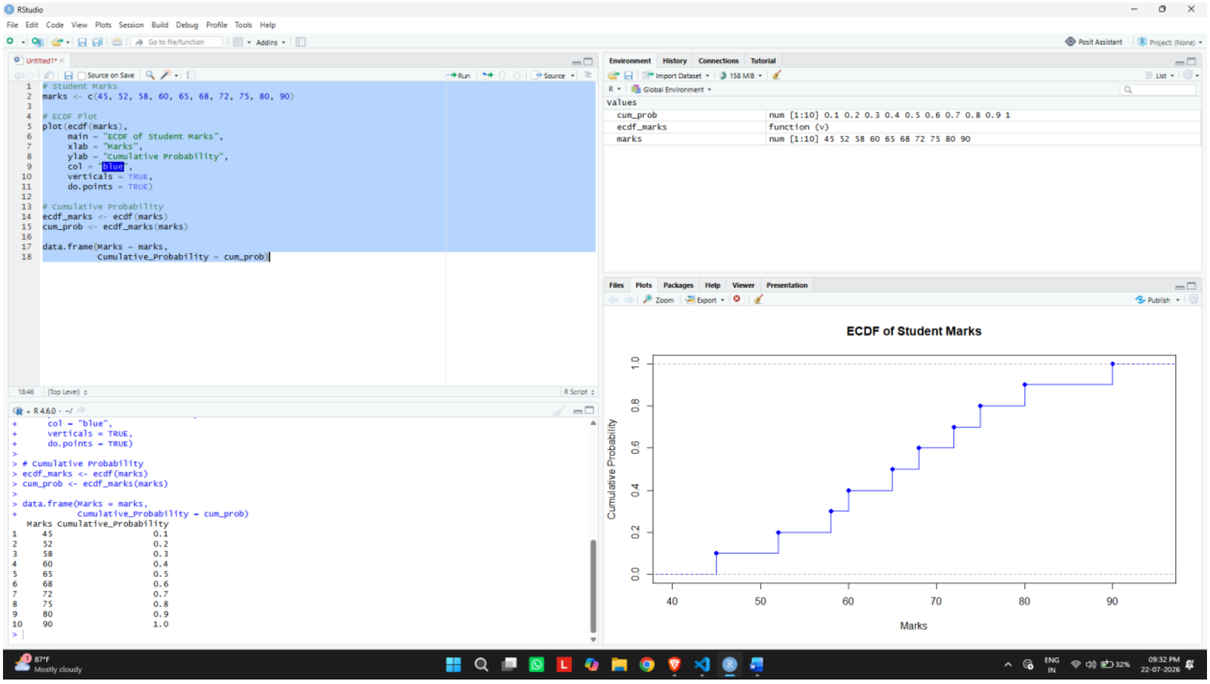
Question 1 – ECDF (Student Marks)

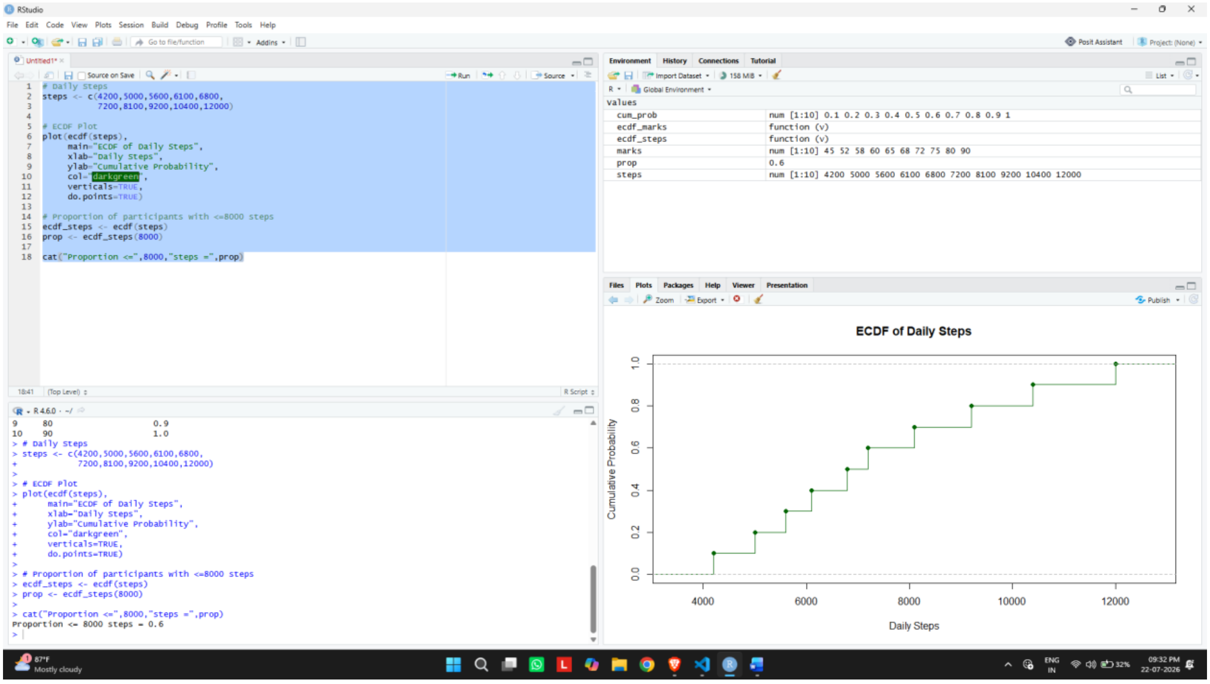
```
# Student Marks
marks <- c(45, 52, 58, 60, 65, 68, 72, 75, 80, 90)

# ECDF Plot
plot(ecdf(marks),
   main = "ECDF of Student Marks",
   xlab = "Marks",
   ylab = "Cumulative Probability",
   col = "blue",
   verticals = TRUE,
   do.points = TRUE)

# Cumulative Probability
ecdf_marks <- ecdf(marks)
cum_prob <- ecdf_marks(marks)

data.frame(Marks = marks,
       Cumulative_Probability = cum_prob)
```



Question 2 – ECDF (Daily Steps)

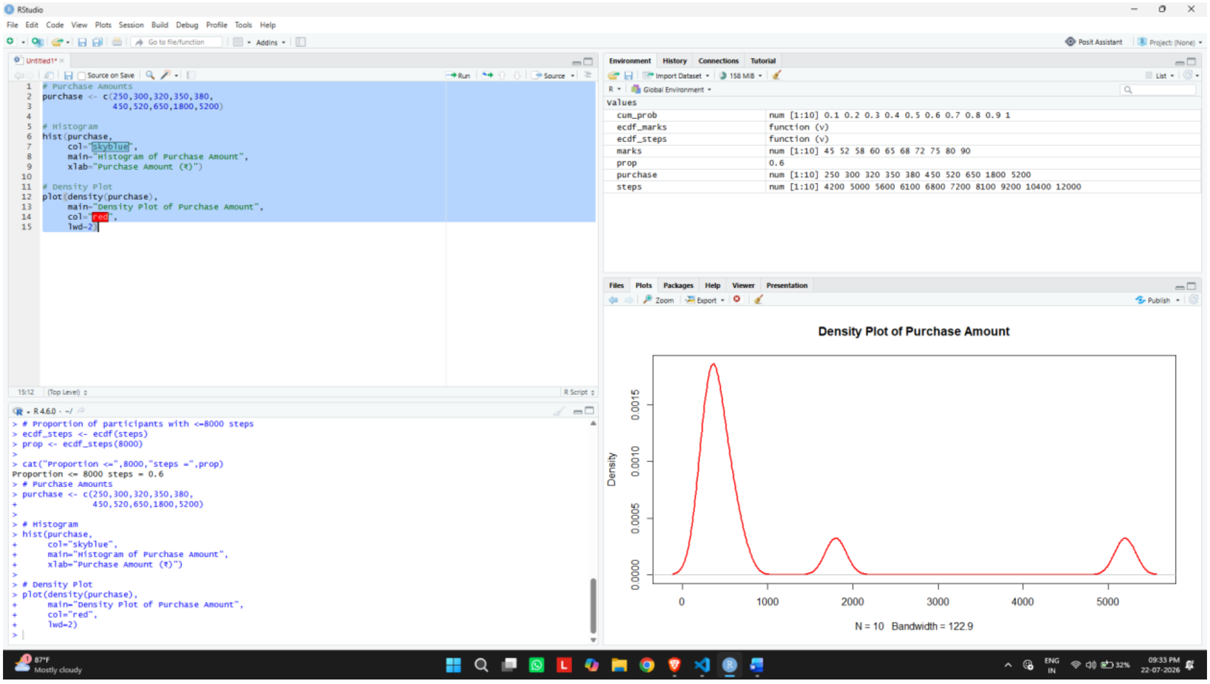
```
# Daily Steps
steps <- c(4200,5000,5600,6100,6800,
       7200,8100,9200,10400,12000)

# ECDF Plot
plot(ecdf(steps),
   main="ECDF of Daily Steps",
   xlab="Daily Steps",
   ylab="Cumulative Probability",
   col="darkgreen",
```

```
    verticals=TRUE,
    do.points=TRUE)

# Proportion of participants with <=8000 steps
ecdf_steps <- ecdf(steps)
prop <- ecdf_steps(8000)

cat("Proportion <=",8000,"steps =",prop)
```



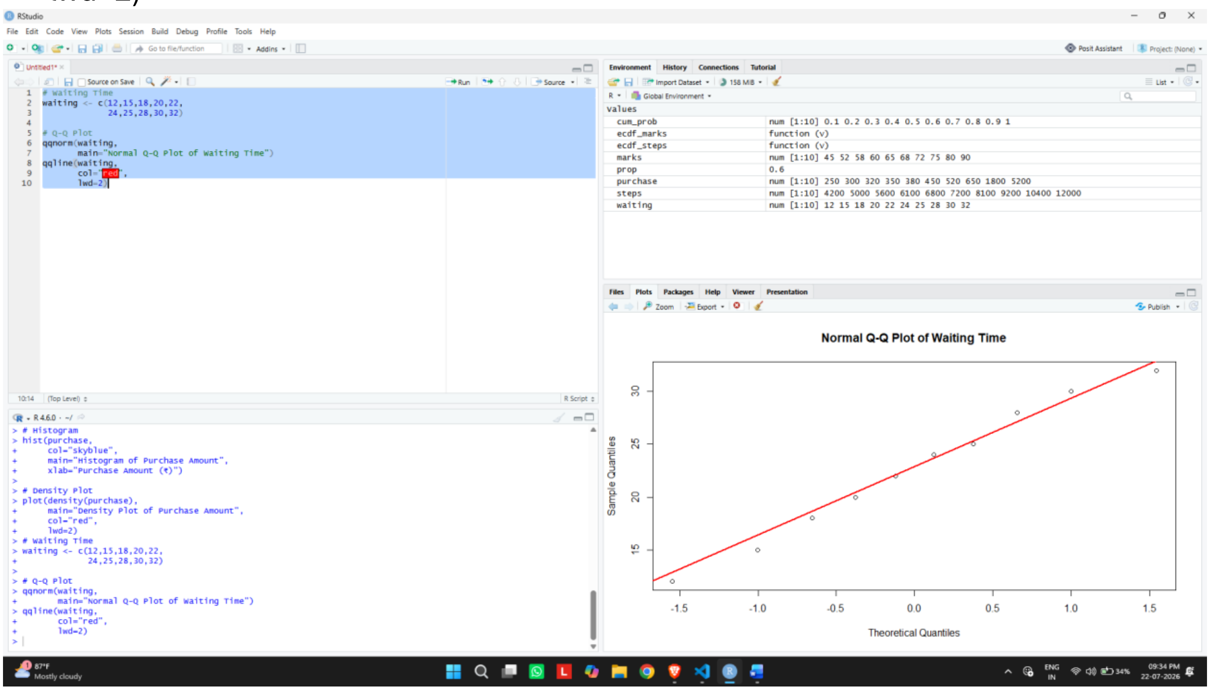
Question 3 – Histogram & Density Plot (Purchase Amount)

```
# Purchase Amounts
purchase <- c(250,300,320,350,380,
          450,520,650,1800,5200)

# Histogram
hist(purchase,
    col="skyblue",
    main="Histogram of Purchase Amount",
    xlab="Purchase Amount (₹)")

# Density Plot
plot(density(purchase),
    main="Density Plot of Purchase Amount",
    col="red",
    lwd=2)
```



Question 4 – Normal Q-Q Plot (Waiting Time)

```
# Waiting Time
waiting <- c(12,15,18,20,22,
         24,25,28,30,32)

# Q-Q Plot
qqnorm(waiting,
     main="Normal Q-Q Plot of Waiting Time")
qqline(waiting,
     col="red",
     lwd=2)
```



Question 5 – Histogram & Q-Q Plot (Monthly Income)

```
# Monthly Income
income <- c(18000,22000,24000,26000,28000,
        30000,34000,42000,85000,200000)

# Histogram
hist(income,
    col="lightgreen",
    main="Histogram of Monthly Income",
    xlab="Monthly Income (₹)")

# Q-Q Plot
qqnorm(income,
      main="Normal Q-Q Plot of Monthly Income")
qqline(income,
      col="blue",
      lwd=2)
```